Nitin Kumar Chidambaram

UNED, Madrid

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Positions

- **Universidad Nacional de Educación a Distancia, Madrid** — Sep 2025 - present
  Ramón y Cajal Fellow (tenure-track)

- **University of Edinburgh** — Oct 2022 - Aug 2025
  Postdoctoral Fellow

- **Max-Planck-Institut für Mathematik, Bonn** — Aug 2020 - Sep 2022
  Postdoctoral Fellow

Education

- **PhD. in Mathematics, University of Alberta** — Sep 2015 - Jul 2020
  Thesis: Aspects of enumerative and categorical algebraic geometry
  Advisors: Vincent Bouchard, David Favero.
- **BTech. Engineering Physics, Indian Institute of Technology Madras** — 2011 - 2015
  Dissertation: Moduli space of string compactifications on K3 surfaces
  Advisor: Prasanta K. Tripathy.

Preprints

1. Chidambaram, Garcia-Failde, Giacchetto, Osuga ‘Refined lattice point counting on the moduli space of Klein surfaces’, arXiv:2605.09526 [math.AG] (submitted)

2. Chidambaram, Dołęga, Osuga ‘b-Hurwitz numbers from Whittaker vectors for $\mathcal{W}$-algebras’, arXiv:2401.12814 [math.AG] (submitted)

Publications

1. Bouchard, Chidambaram, Giacchetto, Shadrin ‘Theta classes: generalized topological recursion, integrability and $\mathcal{W}$-constraints’,
   J. London Math. Soc., to appear

2. Chidambaram, Dołęga, Osuga ‘$\mathfrak{b}$-Hurwitz numbers from refined topological recursion’,
   Math. Ann. 394, 103 (2026)

3. Borot, Chidambaram, Umer ‘Whittaker vectors at finite energy scale, topological recursion and Hurwitz numbers’,
   J. Éc. polytech., Math., 12 (2025), 1503-1564

4. Borot, Bouchard, Chidambaram, Kramer, Shadrin ‘Taking limits in topological recursion’,
   J. London Math. Soc. (2) 112 (2025), no. 3, article no. e70286

5. Chidambaram, Garcia-Failde, Giacchetto ‘Relations on $\overline{\mathcal{M}}_{g,n}$ and the negative r-spin Witten conjecture’,
Invent. Math. 241, 929–997 (2025)
6. Charbonnier, Chidambaram, Garcia-Failde, Giacchetto ‘Shifted Witten classes and topological recursion’,
Trans. Amer. Math. Soc. 377 (2024), 1069-1110
7. Borot, Bouchard, Chidambaram, Creutzig ‘Whittaker vectors for $\mathcal{W}$-algebras from topological recursion’,
Sel. Math. New Ser. 30, 33 (2024)
8. Chidambaram, Favero, ‘Windows for cdgas’,
Adv. Math. 379 (2021), 107553
9. Ballard, Chidambaram, Favero, McFaddin, Vandermolen ‘Kernels for Grassmann Flops’,
J. Math. Pures Appl. (9) 147 (2021), 29–59
10. Borot, Bouchard, Chidambaram, Creutzig, Noshchenko ‘Higher Airy structures, W algebras and topological recursion’,
Mem. Amer. Math. Soc. 296 (2024), no. 1476
11. Bouchard, Chidambaram, Dauphinee ‘Quantizing Weierstrass’,
Commun. Number Theory Phys. 12 (2018), no. 2, 253–303
12. Kleinschmidt, Nicolai, & Chidambaram ‘Canonical Structure of the E_{10} Model and Supersymmetry’,
Phys. Rev. D 91, 085039 (2015)

Grants and Awards

- 2024 Ramón y Cajal Fellowship 2023 — €251,800
- 2020 Alberta Graduate Excellence Scholarship — $12,000
- 2019 Josephine Mitchell Prize 2019 — $10,000
- 2019 University of Alberta Graduate Fellowship — $8,000
- 2019 PIMS Graduate Student Prize, University of Alberta — $5,000
- 2018 Josephine Mitchell Scholarship, University of Alberta — $4,000
- 2017 Eoin L. Whitney Scholarship, University of Alberta — $1,800
- 2016 Josephine Mitchell Scholarship, University of Alberta — $4,000
- 2014 DAAD Fellowship for undergraduate research in Germany — €2,500
- 2009 KVPY[1] fellowship

Teaching

- SMSTC course: Homological Algebra — 2024
- MATH 100: Calculus I — 2020
- MATH 102: Linear Algebra — 2019
- MATH 101: Single variable Calculus — 2017
- MATH 209: Multi variable Calculus — 2016

Teaching Assistant (led discussion sections, gave lectures and graded assignments and exams)

- Teaching Assistant in the Decima Robinson Support Center — 2016-2020

Helped undergraduate students one-on-one with their questions

[1] Kishore Vygyanik Protsahan Yojana is a Govt of India fellowship for basic science research

Supervision

- Georgian Sarghi, Masters in Mathematics, University of Bonn. May 2021 - June 2022

Organization

- “Refinements in Enumerative geometry and physics”, ICTS Bangalore, June 2026.
- Algebra seminar, University of Edinburgh, 2023-2025.
- “Derived, Birational, and Categorical Algebraic Geometry ", BIRS workshop in Banff, November 2021.
- “Derived, Birational, and Categorical Algebraic Geometry (Online) ", BIRS workshop, November 2020.
- “Derived Categories and (Non)commutative Algebraic Geometry ", CMS Winter special session (online), December 2020.

Selected Talks

- “Counting points on the moduli space of Klein surfaces", Colloquium talk, Tata Institute of Fundamental Research, Mumbai, July 2026.
- Mini-course (5.5 hours) “An introduction to topological recursion", University of Nagoya, February 2026.
- “W-algebra representations and Theta classes", CMI-BIRS workshop Geometric and Category theoretic approaches to Conformal Field Theory, Chennai Mathematical Institute, January 2026.
- “Theta classes: integrability, genTR and W-constraints", Enumerative Geometry, Integrability and Topological Recursion, MATRIX, December 2025.
- “Gaiotto State from topological recursion", MaPhyAG seminar, Universidad Complutense de Madrid, October 2025.
- “Theta classes: integrability, W-constraints and topological recursion”, Algebraic geometry and moduli seminar, ETH Zurich, August 2025.
- “Gaiotto state from topological recursion”, Mathematical Physics seminar, University of Nagoya, April 2025.
- “CohFTs from r-th roots and integrability”, Combinatorics and interactions, IMPAN Krakow, March 2025.
- “b-Hurwitz numbers from W-algebra representations”, New Aspects in Topological Recursion, Resurgence and Related Topics, RIMS, Kyoto, July 2024.
- “b-Hurwitz theory, topological recursion and W-algebras ", Geometry Seminar, ICMAT, Madrid, February 2024.
- “Gaiotto vectors from topological recursion", The Sheffield Geometry and Physics Seminar, University of Sheffield, April 2023.
- “On the BGW τ-function", Recent advances on Moduli Space of Curves, Les Diablerets, Switzerland, March 2023.
- “Negative over positive I: The CohFT", New Trends in Mathematical Physics and Geometry session, PRIMA congress, Vancouver, December 2022.
- “Stratified Mukai via VGIT", NSF Workshop on Matrix Factorizations and related topics, University of Notre Dame, November 2022.
- “r-th roots: better negative than positive", Algebra, Geometry and Physics seminar, HU Berlin and MPIM Bonn, November 2022.
- “An introduction to Airy structures and some applications", Algebra Seminar, University of Edinburgh, October 2022.
- Mini-course (4 hours) “An introduction to the moduli space of curves and its intersection theory", TRSalento 2022, Otranto, September 2022.
- “Negative over positive II: Integrability", Geometry and Mathematical Physics Seminar, USTC, Hefei, June 2022.
- “r-th roots: better negative than positive", Colloquium talk, Tata Institute of Fundamental Research, Mumbai, April 2022.
- “Shifted Witten classes and topological recursion", Séminaire de Géométrie Enumérative, Jussieu, November 2021.
- “Kernels for GIT wall crossings", Algebra Seminar, Western University, March 2021.
- “Grassmann flops and Semi-orthogonal decompositions", GLSMs conference (online), August 2020.

- “VGIT for cdgas", Canadian Western Algebraic Geometry conference, University of Saskatchewan, March 2020.
- “$\mathcal{W}$-constraints in enumerative geometry: Higher Airy structures", FRAGMENT seminar, Colorado State University, October 2019.
- “Kernels for singular wall-crossings via cdgas", Geometry and Algebra of Landau-Ginzburg Models, Fields Institute, September 2019.
- “An introduction to Derived Categories in Algebraic Geometry", Graduate Summer School on Homological Algebra of Mirror Symmetry, Fields Institute, July 2019.
- “$\mathcal{W}$-constraints in enumerative geometry ", Alberta Number Theory Days, BIRS, Banff, May 2019.

A full list of talks is available upon request.

Refereeing service

Reviewer for Communications in Mathematical Physics, Dutch Research Council (NWO), Forum of Math Sigma, Geometry & Topology, International Mathematics Research Notices, Inventiones Mathematicae, Journal of the European Mathematical Society, Journal of the London Mathematical Society, La Matematica, Selecta Mathematica, Transactions of the AMS, US-Israel Binational Science Foundation.